J-space Phase 3 — mechanism, generalization, paper hardening

working handout

branch `interp_jspace_part2` · run root `phase3_20260729` · 2026-07-30, VM10 late block (pre-freeze refresh)

Tier legend. Every number carries its evidence tier: PHASE2-CONFIRMATORY results are frozen and immutable; PHASE3-DEVELOPMENT results guide design and are NOT confirmatory claims; METHODS results validate instruments. Nothing here exists outside `reports/evidence_events.jsonl`.

1 Where Phase 3 stands

Phase 2 closed with both preregistered primaries rejecting (HP1 task-shape contrast $-0.504\,[-0.720, -0.295]$; HP3 Qwen J-specific tail $+0.279\,[+0.205, +0.361]$, replicating held-out at $+0.297$) PHASE2-CONFIRMATORY. Phase 3 attacks the four gaps that block the strong manuscript sentence: the thin two-hop leg, the label-vs-span protection asymmetry, the missing prose exact control, and full clean-room reproduction.

- **§4.1a overlap mining** `p3-overlap-mining-v1` — done.
- **N8 Level 1 clean room** `p3-n8-level1-repro-v1` — done, all four primaries reproduce to ≤5e-4.
- **§4.1b span audits** — done on ALL THREE primaries + cross-model synthesis `p3-span-audit-cross-model-v1`.
- **Workstream C prose exact control** — COMPLETE on all three primaries (`p3f04`).
- **Banks F/S** — 72/72 families authored, validated, source-verified; G5-scored ×3.
- **FROZEN + BLOCK B COMPLETE**: tag `jspace-phase3-freeze-v1`; primary cells ×3, locked analysis, replication side, N8 Levels 1–3 all banked.

2 THE PHASE 3 PRIMARY RESULTS

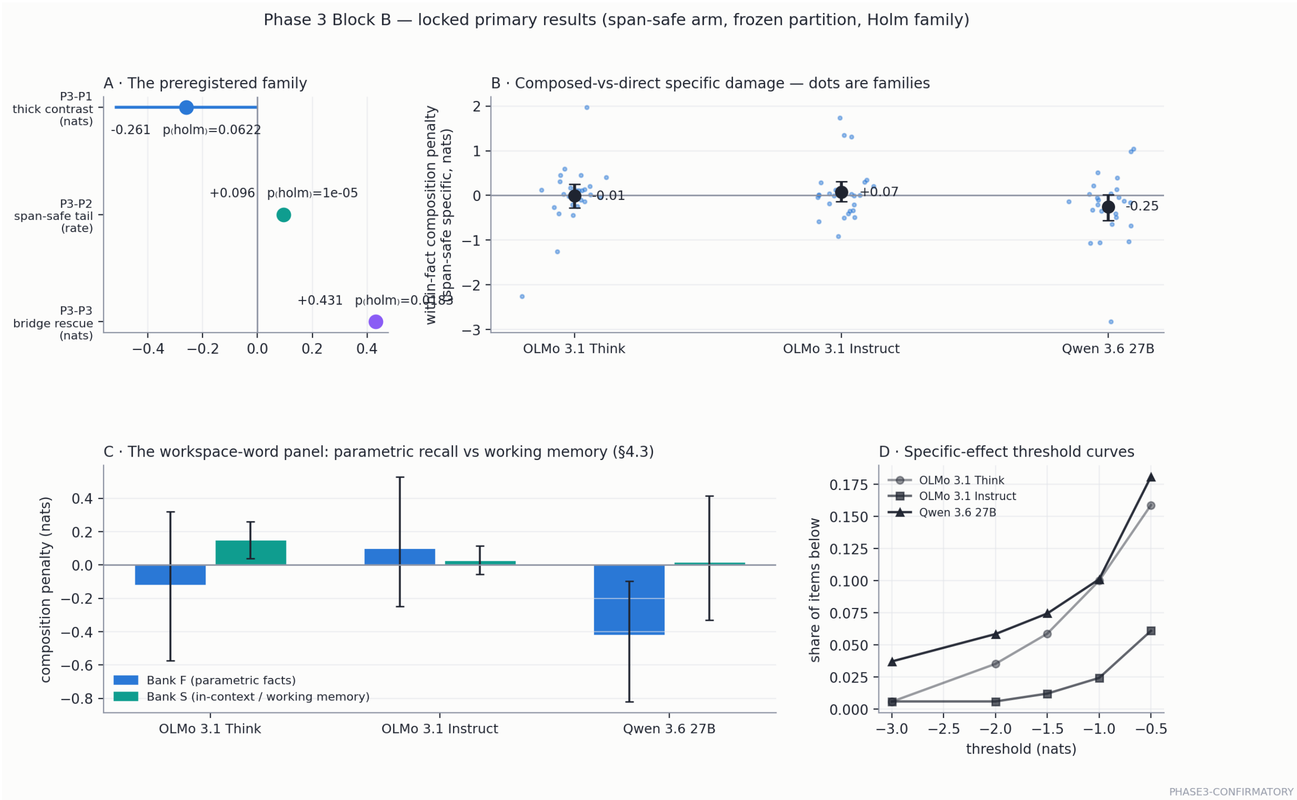
phase3-confirmatory

Locked analysis `p3-locked-analysis-v1` (run once, raw parquets, after the last cell; span-safe primary arm; frozen partition seed 85670; Holm family):

primary	estimate	p_{Holm}	verdict
P3-P2 span-safe tail, Qwen	+0.096 tail excess (188 items / 26 fam)	∼1e-5	**REJECTS**
P3-P3 bridge rescue, Qwen	+0.431 nats (94 composed items)	0.018	**REJECTS**
P3-P1 thick contrast, Qwen−OLMo	-0.261 nats $[-0.52, -0.01]$ (17 fam)	0.062	directional, under disclosed MDE

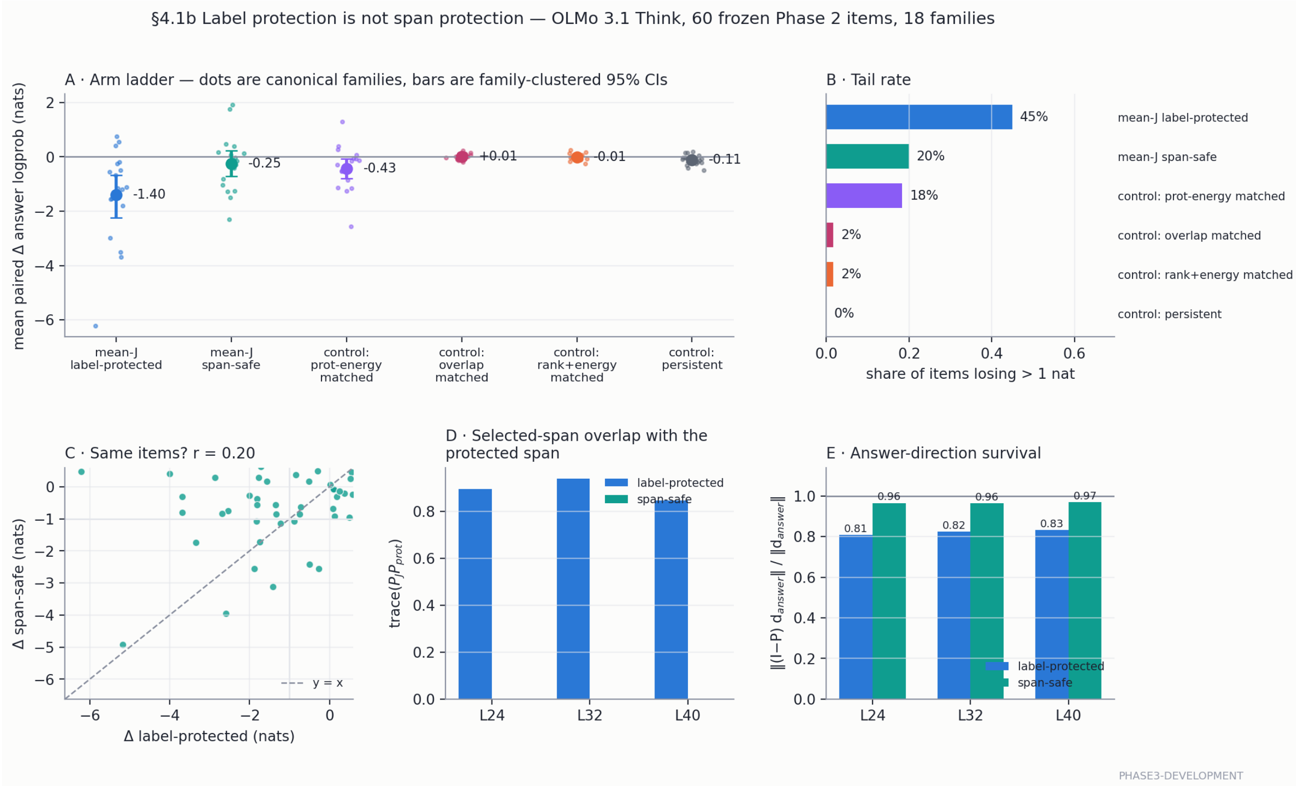
(1) The J-specific heavy tail *survives full span-safe protection* at confirmatory tier — and **REPLICATES on held-out families** (+0.102, $p \approx$1e-5, 28 fam) `p3-replication-analysis-v1`. The campaign's central specificity claim now rests on the leak-proof arm, at ∼1/3 of the label-protected size, exactly as the dev decomposition predicted. **(2) Bridge mediation is confirmed** on the magnitude-blind-chosen model. **(3)** P3-P1 is paper-shaped in direction with agreeing sensitivities, under the preregistered power disclosure — the estimation targets carry the story: Think≈Instruct on the thick bank ($+0.01\,[-0.45, +0.39]$); and the §4.3 workspace-word test says Qwen's contrast does NOT appear on Bank S (+0.02, wide CI) — by the pre-stated rule, the honest noun is **"knowledge-access channel"**, not working-memory workspace. (Think's Bank-S term is CI-clean *positive*, $+0.15\,[+0.04, +0.26]$ — flagged for mechanism.)

Reproduction ladder discharged: N8 L2 sentinels bit-exact ×3 (worst dev 0.0); **L3 full 184-item Qwen cell bit-exact** `p3-n8-level3-qwen36-27b-v1`.



3 §4.1a — Selection pressure is a model property, not a damage predictor phase3-development

Protected-row selection pressure (clean top-k rows the J selection would have taken) does not predict per-item damage in any model (ρ 0.00–0.11, CIs cross 0), *but* it is a 12× model property: median blocked rows/position Think 1.33, Instruct 1.22, Qwen **0.10**. Accessibility signs flip between clusters: $\rho(\text{clean rank}, \Delta)$ $-0.30/-0.31$ on the OLMo pair (CI-clean) vs +0.24 on Qwen. `p3-overlap-mining-v1`, fig. `p3f01`.



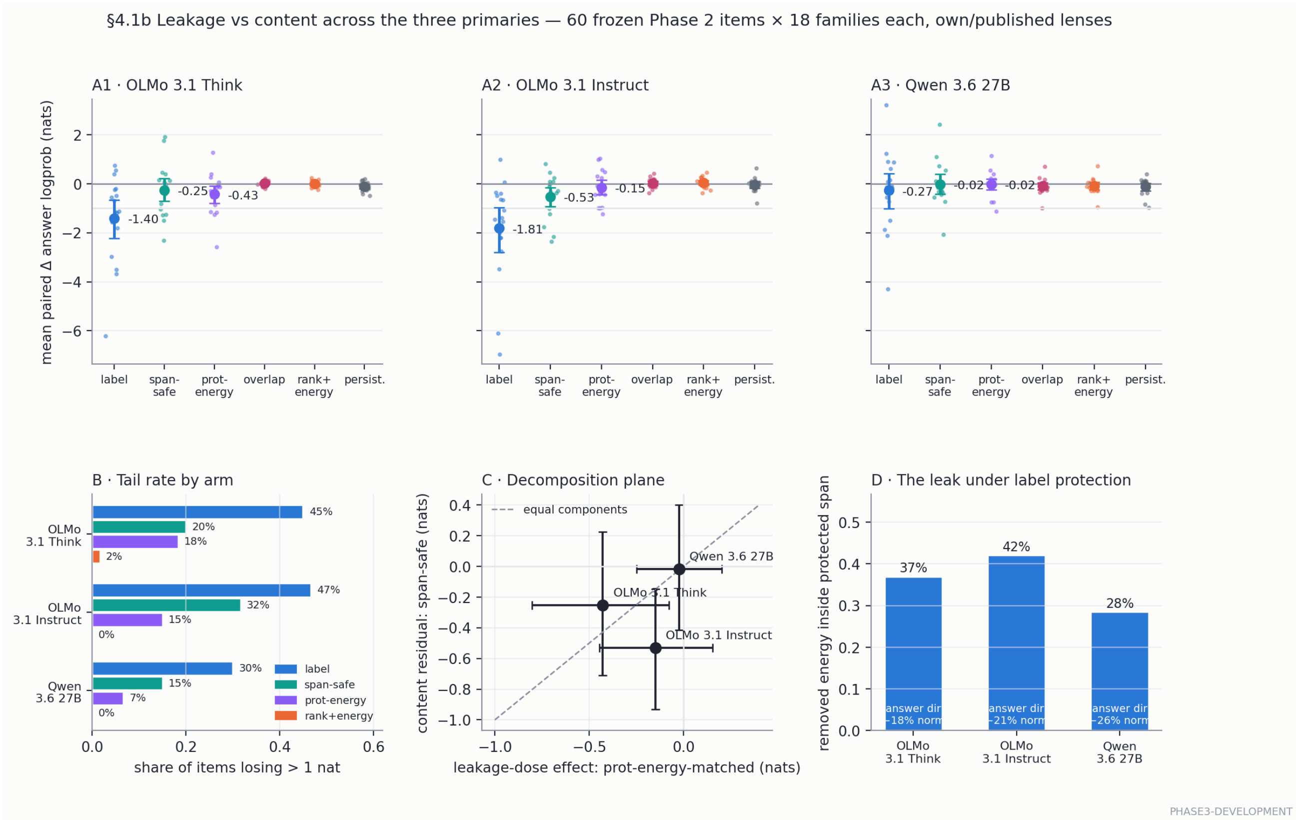
4 §4.1b — Label protection is not span protection

phase3-development

The paper-faithful label-protected arm *leaks into the very span it declares protected*, on every model (fig. above: OLMo 3.1 Think own lens, 60 frozen Phase 2 items). Span-safe protection zeroes the overlap exactly (its own positive control) and the effect decomposes:

family-weighted Δ (nats) · tail@−1	Think	Instruct	Qwen
label-protected	−1.40 [−2.24, −0.67] · 45%	−1.81 [−2.81, −0.97] · 47%	−0.27 [−1.01, +0.42] · 30%
span-safe	−0.25 [−0.71, +0.23] · 20%	**−0.53** [**−0.93**, **−0.15**] · 32%	−0.02 [−0.41, +0.40] · 15%
prot-energy-matched	**−0.43** [**−0.80**, **−0.08**] · 18%	−0.15 [−0.45, +0.16] · 15%	−0.02 [−0.25, +0.20] · 7%
span-safe own control	−0.005 · 2%	+0.01 · 3%	−0.08 · 2%
specific span-safe tail	+18pp	+28pp	+13pp
r(label, span-safe) per item	0.20	0.29	**0.75**
removed energy inside protected span	37%	42%	28%
answer-direction norm lost	18%	21%	26%

Reading. (1) The leak *geometry* is universal — every model deletes protected-span energy under label protection. (2) The *behavioral* leakage component is essentially a Think phenomenon (prot-energy control reproduces ∼1/3 of its label mean, nil on Qwen). (3) The content residual is largest and CI-clean on Instruct. (4) A J-specific tail survives full span protection on all three (own controls ≤3%). (5) On Qwen, span-safe *rescales* the same items ($r = 0.75$); on OLMo it *reallocates* — so the decomposition must be carried as two components, never one number. Phase 2's HP3 transfers nearly intact on Qwen, as §4.1a's selection-pressure contrast predicted. p3-span-audit-{olmo31-think-v2,olmo31-instruct-v1,qwen36-27b-v1} fig. p3f03.



⇒ meanJ_span_safe is the P3-P2 primary arm; prot_energy_matched its preregistered comparator. Scope: development tier; the sharp control was designed after seeing the Think overlap numbers (fine for arm selection, disqualifying for confirmatory claims); confirmatory versions run on the thick bank post-freeze.

5 Instrument notes

methods

- **N8 Level 1 passes**: a narrative-blind agent regenerated the full Phase 2 primary table from frozen parquets in a clean output root; all four quantities match to $\leq$5e-4. Levels 2–3 (sentinel subsets, one full GPU cell) remain open as release gates. `p3-n8-level1-repro-v1`
- **BOS units on Qwen**: `ScoringSession` demanded a `bos_token_id`; Qwen has none, and `jlens.from_hf(force_bos=True)` is a no-op there — every frozen Phase 2 Qwen artifact is in *native* units. The session now scores native exactly when no BOS exists and records the applied convention (commit `70d8eae`).
- **Units verified bit-exactly**: all three span audits reproduce frozen N6 per-alias baselines exactly (60/60 items per model). Where an audit baseline differs from N6's `lp_canonical`, the audit's alias choice named a different case variant — same machinery, same units.
- **Report CI correction**: the Think §4.1b section's first published label/span-safe/prot-energy CIs matched no registered artifact (transcribed from a mid-audit dev render). REPORT_PHASE3 now quotes the registered `p3-span-audit-figure-olmo31-think-v2` values (as does the table above); point estimates, tails, and conclusions unchanged.

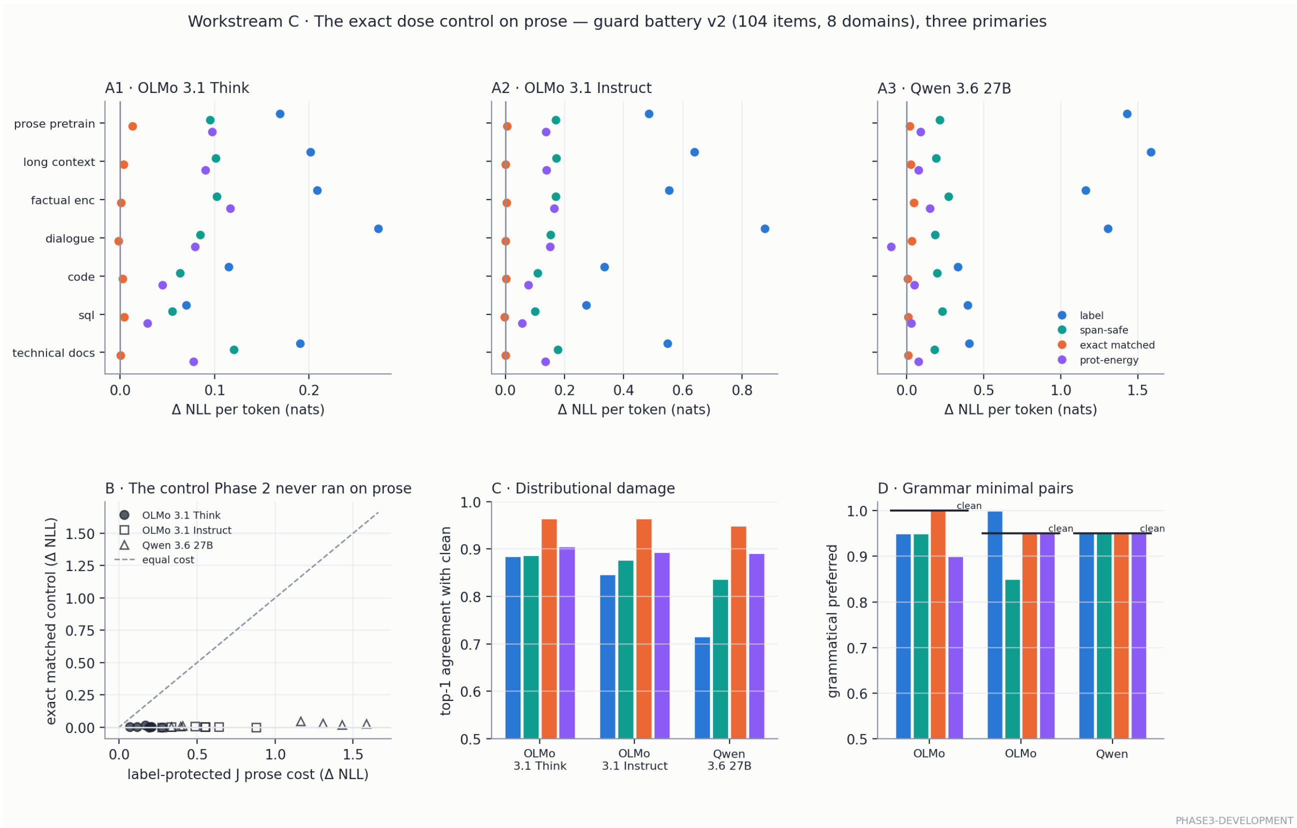
6 Workstream C — the exact control on prose: COMPLETE

phase3-development

Phase 2's prose guard skipped the primary matched control; §7.1 made it mandatory before any "selective" wording (R4). Guard battery v2 `p3-guard-battery-v2` (104 items / 8 source-pinned domains, wikitext TEST held out from every lens-fit corpus), six arms per item including the profile-consuming exact instant rank+energy control. `p3-prose-grid-{olmo31-think,olmo31-instruct,qwen36-27b}-v1`, figure `p3f04` + `p3-prose-grid-figure-v1`.

mean Δ NLL / token	Think	Instruct	Qwen
label-protected J	+0.175	+0.530	**+0.945**
span-safe J	+0.089	+0.150	+0.209
exact instant matched	**+0.003**	**+0.002**	**+0.021**
prot-energy matched	+0.076	+0.123	+0.052

(1) The prose cost is J-content everywhere, never dose (exact control ≈0 on all three). **(2) The prose ladder INVERTS the task ladder and tracks the §4.1b projector-overlap ladder** (Qwen > Instruct > Think ↔ 1.37 > 0.94 > 0.89): on prose every position leaks, and span-safe removes 72–78% of the label cost on every model. **(3) No model earns "selective" wording at this tier**: the §7.4 standardized index is +1.25 to +1.57 — prose damage exceeds task damage in standardized units on ALL three, so the R4 sentence stands for every primary ("nonspecific J-channel vulnerability cannot be excluded"), sharpened: the vulnerability is content-specific and predominantly output-span leakage. Grammar preference survives every arm (85–100%).



7 Banks F and S — 54 of 72 families

phase3-development

- **Bank F v5** `p3-bank-f-v5`: **164 bundles / 38 families**, every bundle direct/composed/bridge-supplied + rotated counterfactual, functional first hops with per-family ambiguity notes, cross-phase dedup against every Phase 2 (bridge, answer) pair AND bare answer (858 burned answers), one (bridge, answer) pair per bank, distinct templates per family. **164/164 verified** against a pinned reference.
- **Bank S v2** `p3-bank-s-v2`: **80 bundles / 16 synthetic template families** (lookup, binding, path, distractor, reversible-mapping types per §5.2) — the working-memory bank whose composed−direct contrast is preregistered as Family B's first member (addendum §4.3).
- **Verification pivot** `p3-wiki-reference-v4`: the Drive `wikimedia/wikipedia 20231101.en` snapshot is hub-authentic (shard sha256 verified) yet *missing the plain articles of major entities* — 'Thailand' is absent from all 6,407,814 rows while "Thailand at the 1992 Summer Olympics" exists; the release evidently drops its largest pages. Bank verification therefore runs against freshly fetched wikitext pinned by *per-page revision ids* (strictly stronger provenance than a dump date). 248/248 pages fetched; every hop of every shipped bundle is stated on its source page.

- **Complete**: 48 F + 24 S = 72 families / 324 bundles; G5 $\times 3$ (972 items each). Two grading-instrument catches from reading failed generations: `normalize` deleted whitespace runs ("the\nBaht"→"thebaht", regression-tested fix), and the inclusion predicate moved to *answer contained in the 8-token greedy continuation* (endpoints are teacher-forced lp; prefix rate kept as a reported column). Cohorts: 73.5/74.2/77.9% capable; three-model intersection **104 facts / 34 families**. The R2 $\geq$25/side intersection floor is infeasible with this bank (the Phase 2 burn list forces less-famous instances exactly where three-model capability is scarce) — floor amended to measured feasibility ($\geq$16/side), *disclosed* in prereg §5. Split preview `p3-family-split-preview-v1`: 36/36 families, F 24/24, intersection 16/18, imbalances $\leq$0.35.
- **P3-P3 = Qwen** by the pinned tie-break (both gates identifiable at 0.85 coverage; ties→Qwen, rule pinned before the Qwen gate ran). **Power** `p3-power-sim-v3`: P3-P2 MDE $\approx$0.2 tail-points at 30–36 families (anchor +0.279); P3-P1 powered only for $\geq$0.3–0.5-nat effects (disclosed scenario table); SESOI 0.15 nats.

8 Paper tree

Sections 1–3 (problem, assay repair, Phase 2 matrix) are drafted from frozen material only, with evidence ids inline and the §0.2 calibrated sentence verbatim (`paper/01--03*.md`). Writing them surfaced no overclaims to withdraw; the assay-repair section doubles as the methods-contribution narrative.

Standing rules: Phase 2 frozen and immutable · producers refuse dirty trees · supersede, never edit · tier watermarks everywhere · the contradiction heuristic (it caught the BOS guard and the wikipedia-release defect this block) · prohibited claims per nextsteps §4.5.